# Pell Congruences, Shared Positivity, and Smaller Diophantine Consistency Equations

Research contribution prepared for Vladimir Reshetnikov
AI-assisted derivation and independent internal audits

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**Abstract**

We give explicit fixed-coefficient polynomials whose absence of integer solutions is equivalent, by ordinary mathematics formalizable in ZFC, to syntactic consistency of ZFC. Relative to the pinned wave-10 ProveIt snapshot, the principal new integer tradeoffs include a quartic in 37 unknowns, an octic in 26, degree 10 in 24, degree 16 in 22, degree 20 in 20, degree 32 in 18, and degree 88 in 17. The main new compression replaces two auxiliary quotients by one coordinate using a congruence which the preliminary Pell rank argument already proves. This avoids a circular appeal to the exact rank which those quotients are intended to certify. A parity-forced change of the first Pell root, an exact transport projection, and compatibility between existing determinant and residue guards yield further reductions. We also give eleven-positive-variable degree 269 and twelve-positive-variable degree 78 equations. A separately certified nine-positive-variable companion trades more arithmetic gates for lower degree. All coefficients, literal polynomials, builders and finite certificates accompany the article. These are improvements to an inspected collection, not a proof of global minimality or a new machine-independent calibration. The native compiler, specified machine/refutation equivalence and named Pell rank theorems remain explicit dependencies; no new end-to-end proof-assistant certificate is claimed.

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# 1 Results, comparison and logical meaning

The comparison is frozen at ProveIt commit `cf4a5218a5a4e4358d3f5314151b6b05d4aa384e`. This is the README reconciliation after wave 10, including packages 68–75. The repository's own review lists 38 unrestricted integer variables at degree four, 32 at degree six, 27 at degree eight, and 26 at degree sixteen. It also retains an older 24-integer degree-1936 form. These are domain-qualified results: the older sextic in 25 *positive* variables and octic in 20 positive variables do not describe unrestricted-integer zero sets [1, 2].

The following new forms all use exactly the stated domain, actual fixed integer coefficients, and one final polynomial equation.

| Domain | Unknowns | Exact degree | Residuals | Main mechanism |
|---|---|---|---|---|
| $\mathbb{Z}$ | 37 | 4 | 24 | Determinant plus residue orientation |
| $\mathbb{Z}$ | 31 | 6 | 18 | Monic projection |
| $\mathbb{Z}$ | 26 | 8 | 13 | Monic projection |
| $\mathbb{Z}$ | 24 | 10 | 11 | Centered root and transport projection |
| $\mathbb{Z}$ | 23 | 12 | 11 | Shared positivity |
| $\mathbb{Z}$ | 22 | 16 | 10 | Shared positivity and transport projection |
| $\mathbb{Z}$ | 20 | 20 | 8 | Retain the rank as one coordinate |
| $\mathbb{Z}$ | 19 | 28 | 7 | CRT and transport projection |
| $\mathbb{Z}$ | 18 | 32 | 6 | Further product projection |
| $\mathbb{Z}$ | 17 | 88 | 5 | Eliminate the remainder coordinate |

Here "residuals" means that the single polynomial is the sum of their squares. The six-residual, eighteen-variable form is displayed in full in Section 6. Named polynomial registers in that display are abbreviations, not additional quantified variables.

The integer results are accompanied by the following positive-domain tradeoffs. An operation is one live binary addition, subtraction or multiplication; squares and multiplication by constants

are charged. Fixed numerals and reuse of a previously computed register are free. This is an arithmetic-circuit convention, not bit complexity.

| Construction | Positive unknowns | Exact degree | Operations |
|---|---|---|---|
| Wave-10 source 70 | 11 | 325 | 113 |
| New saturated transport | 11 | 269 | 111 |
| Wave-10 source 69 | 12 | 82 | 102 |
| New adaptive saturated merger | 12 | 78 | 102 |
| Wave-10 source 71 | 9 | 1,379,730 | 478 |
| New full-radical majorant | 9 | 1,158,228 | 726 |

The nine-variable companion improves degree but increases evaluator size. It does not replace the separate 443-operation local nine-variable target reported by the active research. The eleven-variable result improves both degree and the stated operation count.

### 1.1 The exact metamathematical assertion

Fix the recursively specified ZFC calculus used by the repository's machine calibration and let $\mathrm{Ref}_{\mathrm{ZFC}}(p)$ mean that $p$ codes a finite derivation of a contradiction. Put

$$\mathrm{Con(ZFC)} \;\equiv\; \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

The construction establishes, with the explicitly named inherited interfaces, the arithmetic theorem

$$\mathrm{ZFC} \vdash (\exists \boldsymbol{x} \in \mathbb{Z}^n\ P(\boldsymbol{x}) = 0) \longleftrightarrow \exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p). \tag{1}$$

Negating both sides gives the requested no-solution equivalence. Positive variables replace $\mathbb{Z}^n$ in the positive companions. No coefficient is defined by deciding consistency, waiting for a refutation, or consulting an oracle. No assertion that ZFC is consistent is needed in proving the biconditional.

Ordinary mathematical formalizability and a supplied formal proof are different deliverables. The new arguments below use integer arithmetic, finite recurrences, classical number theory and induction, all available in ZFC. This package does not contain a newly kernel-checked Lean/Rocq proof or a serialized derivation of (1) in the repository's object-language calculus. Its finite computer checks certify coefficients and algebra, not the unbounded semantic interfaces.

### 1.2 What is new and what is reused

The determinant guard is package 71's construction; residue orientation is package 70's; the nearest-square joint positivity lemma is package 68's. We prove their simultaneous compatibility, then add a lossless CRT parametrization at the correct stage of the signed Pell argument. The elementary CRT identity is not asserted to be a new number-theoretic theorem. Its use to remove these two source quotients, with a proof that the needed congruence is available before auxiliary rank recovery, is the new structural reduction. Replacing the slack coordinate by the native $C$ coordinate also makes the transport row monic in $F$, enabling one more exact projection. Centering and further projection produce the new literal equations and their degree certificates. No external priority claim beyond the inspected sources is made.

The main integer family uses neither Jones's universal polynomial nor Sun's ten-integer reduction. It still uses the calibrated native encoding of K215. A separate analysis of direct proof-checker and finitely axiomatized class-theory routes appears in Section 11; those alternatives are not counted as completed new equations here.

## 2 Fixed coefficients and the inherited source

The integer forms use the complete source-34 tuple

$$(\beta, K_0, M_C, M_F), \qquad \beta = 2^{1064921} - 1,$$

with $M_F$ already shifted by $\beta$. Two additional fixed scales are

$$A_F = 2^{1113}, \qquad A_C = 2^{666104}. \tag{2}$$

These six numbers are constants of the polynomial, not witness variables. The package stores all six actual integers as gzip-compressed minimal unsigned little-endian byte strings, with hashes of the decompressed integers. Their total uncompressed size is 615,883 bytes. This is a real coefficient-size cost, not hidden in the variable count.

### 2.1 A finite recipe, not a parameter family

The retained `coefficients/source34/build_coefficients.py` defines the four native constants by a finite recipe from the literal 430 transitions of K215. It derives and verifies the ordered 943-tile table. With $\rho = 2^{121}$, block length 8801, and $B = \rho^{8801}$, the recipe constructs finite selector/copy weights and a finite convolution polynomial $D_C$. It sets

$$\begin{aligned} \beta &= B - 1, & K_0 &= D_C + B\rho, \\ M_C &= \text{the specified low mask}, & M_F &= \text{the specified high mask} + B - 1. \end{aligned}$$

The masks, tile order, support positions and every finite sum are literal in that short builder and its retained coefficient proof. The supplied payloads are checked against the builder, including the mask shift. Consequently the displayed formulas specify one closed polynomial once these attached fixed numerals are substituted. The builder never runs the blank machine to completion. The compact source-44 tuple used later for positive companions is kept separately and is never mixed into these integer coefficients. The nine-positive-variable companion uses the complete source-31 tuple instead: its transport shift is $D_R = \rho^{42}$, whereas source 34 uses $D_R = \rho$. Equality of their $\beta$ values does not make the other coefficients interchangeable.

### 2.2 The source theorem we import

Packages 64 and 70 supply the signed native/Pell source used here [3, 6]. The following is the exact logical boundary. The source has arithmetic identities which, after eliminating its monic registers, read

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, & X &= qW, \\ Y &= s_0 q^3, & E &= XY, & a &= (X+1)Y, \\ H &= 4a + 3, & \Delta &= (a+2)^2 - 1, & k &= \eta + \zeta, \\ c &= kY + \eta, & P &= k + E(c - \eta), & D &= X + ac + H\gamma, \\ I &= i_0 c^2, & I^2 &= \Delta(f^2 - 1), & V &= fo - c = jc - R. \end{aligned}$$

The remaining abbreviations are

$$R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \qquad k = R + 1 + hE.$$

Here $s_0, i_0$ are odd, $W$ is a nonzero multiple of four, and the source enforces the common-sign requirements

$$F\alpha > 0, \quad FC > 0, \quad \eta\zeta > 0.$$

Its four remaining norm/transport equations are

$$\tau_0^2 - P^2 + Pk - 1 = 0, \tag{3}$$
$$D^2 - \Delta c^2 - 1 = 0, \tag{4}$$
$$I^2V^2 - (I^2 - 1)y_0^2 - 1 = 0, \tag{5}$$
$$(K_0 + W)C - F - z(q - 1) = 0. \tag{6}$$

Package 70 removes the separate $X = 2u_0^2$ witness by imposing $J = 3J_3 + 1$ and $W = 4(3w_3 + 1)$. Its signed power theorem first proves $X = \pm 2^R$ with odd $R$; then $X \equiv 2 \pmod 3$ forces the positive sign. Completing the source gives a finite refutation, and any finite refutation gives canonical source witnesses.

One part of this proof must be distinguished from its conclusion. Before using either congruence for $V$ or the auxiliary norm (5), the first/main norms, the index relation and the ordinary strong equation yield

$$|c| = \psi_b(p), \quad |f| = \chi_b(m), \quad p \mid m, \quad c \neq 0, \quad f \neq 0, \qquad b = |a + 2| \geq 2. \tag{7}$$

The proof appears in package 70, Sections 4.1–4.2, and the retained ordinary strong-rank proof. The exact rank $p = R$ is obtained only later. This distinction is essential in Section 4.

## 3 Compatible sign reductions and a 37-variable quartic

For integers define

$$\mathrm{Pos}(x, y, z) = 1 + x^2 + y^2 + z(z + 1).$$

It is always positive and represents every positive integer. For the latter assertion apply the three-square theorem to $4n - 3$; one of the three squares is odd and the other two are even. The familiar representation of nonnegative integers by $x^2 + y^2 + z^2 + z$ is also formally documented in [10].

**Lemma 3.1** (Determinant sign recovery)**.** *For fixed $A_F, A_C \geq 2$, put*

$$U = A_F F, \quad W_C = A_C C, \quad q = F + \alpha + C, \quad G = UW_C - (q - U - W_C)^2.$$

*If $G > 0$, then $F, C, \alpha$ have one common nonzero sign.*

*Proof.* The inequality implies $UW_C > 0$ and $|q - U - W_C| < \sqrt{UW_C} \leq (|U| + |W_C|)/2$. Thus $q$ has their common sign and $|q| > (|U| + |W_C|)/2 \geq |F| + |C|$. Subtracting $F + C$ gives the same strict sign for $\alpha$. No sign of $q$ has been assumed. □

**Proposition 3.2** (Simultaneous canonical completeness)**.** *The determinant guard with (2) and the two affine residue restrictions of package 70 are simultaneously satisfiable on the canonical source witnesses of every finite refutation.*

*Proof.* The source-34 canonical family has $q = 2^t$, $t = 1064921N$ odd, $R \equiv 3 \pmod 4$ and $R > t$. Its original coordinates satisfy

$$J = \frac{2^t - 1}{2^{1064921} - 1}, \qquad w_{\mathrm{old}} = 2^{R-t-2}.$$

Therefore $J \equiv 1 \pmod 3$ and $w_{\mathrm{old}} \equiv 1 \pmod 3$, so $J = 3J_3 + 1$ and $w_{\mathrm{old}} = 3w_3 + 1$ have integer inverse coordinates.

Package 71 proves, for this same rank-tuned family,

$$\frac{q}{2} \le A_F F < \frac{4q}{5}, \qquad \frac{q}{2} \le A_C C < \frac{3q}{5}.$$

Its proof is retained in the archive. The highest occupied $C$ digit is at exponent 3296 in radix $2^{121}$; all lower support is at most 2195. The leading $F$ digit at exponent 8791 has coefficient 105534389824824194647382298625. The rank controller changes a dummy at exponent 1063, below these leading positions, and the carry bounds cover every permitted dummy change. Thus the estimates are compatible with the parity/rank tuning, rather than results for two unrelated witness families.

Write $U = q/2 + x$, $W_C = q/2 + y$. Then $0 \le x < 3q/10$, $0 \le y < q/10$, and

$$G = (q/2)^2 + (q/2)(x + y) - x^2 - xy - y^2 > 3q^2/25 > 0.$$

Choose a ternary Pos representation of this positive integer. □

Consequently package 70's two guards on $F\alpha$ and $FC$, with six private coordinates, can be replaced by $G = \mathrm{Pos}$ with three. Soundness restores the two former guards *before* invoking the signed power theorem. Completeness uses the one compatible family just proved. The result is a quartic in 37 integers, the sum of 24 quadratic residuals. Its coefficient of $\tau_0^4$ is one.

The direct-coordinate alternative replaces only $X$ by $3X_3 - 1$ and also has 37 unknowns. It is included as `quartic37_direct.json`; it has a shorter expanded expression. The affine-$J, w$ version is used for the monic projections below, because it retains a monic definition of $X$.

## 4 The new CRT compression

**Lemma 4.1** (One parameter for two source congruences)**.** *Let $c, f$ be nonzero integers with $c \mid f^2 - 1$. For every integer $R$, the simultaneous congruences*

$$V \equiv -c \pmod f, \qquad V \equiv -R \pmod c$$

*hold exactly for the integers*

$$V = fcT - f^2R - c, \qquad T \in \mathbb{Z}. \tag{8}$$

*For fixed $c, f, R$, this parametrization is one-to-one.*

*Proof.* Since $f^2 \equiv 1 \pmod c$, we have $\gcd(f, c) = 1$. Write $V = fo - c$. The second congruence implies

$$c \mid f(o + fR) = V + c + f^2R,$$

so $c \mid o + fR$. Set $T = (o + fR)/c$. Conversely, from (8) take

$$o = cT - fR, \qquad j = fT - 1 - \frac{(f^2 - 1)R}{c}. \tag{9}$$

The second expression is an integer by hypothesis, and these give $V = fo - c = jc - R$. The formulas are inverse. Signs do not enter the proof. □

**Lemma 4.2** (The CRT hypothesis is available early)**.** *Every tuple satisfying the preliminary signed source equations used in* (7) *satisfies* $f^2 \equiv 1 \pmod{c^2}$.

*Proof.* The Pell identity is $\chi_b(m)^2 - 1 = (b^2-1)\psi_b(m)^2$. Since $p \mid m$, divisibility of the Pell sequence gives $\psi_b(p) \mid \psi_b(m)$. Use (7) and square the divisibility. This does not use the auxiliary norm, either quotient $o, j$, or the later conclusion $p = R$. □

**Theorem 4.3** (Lossless quotient elimination)**.** *In the residue-oriented, determinant-guarded source, the two rows* $V = fo - c$ *and* $V = jc - R$*, and their private coordinates* $o, j$*, may be replaced by the one row* (8) *and one coordinate* $T$ *without changing existence of an integer solution.*

*Proof.* For soundness first recover the signs and preliminary rank facts from the equations which do not use $o, j, V, y_0$. The preceding lemma supplies the hypothesis of Lemma 4.1. Restore $o, j$ by (9); the rest of the source, including its auxiliary norm, is now literally available. Its final rank/calibration proof applies. For completeness take any source solution and use the forward CRT formula for $T$. All other coordinates stay unchanged. □

There is no stronger requirement $f \equiv \pm 1 \pmod{c}$. Nor do we impose $f^2 \mid V + c$: that would retain only a subprogression and need a new completeness argument. The coefficient of $cT$ in (8) is $f$, and this distinction is what makes the reduction lossless. The only displayed division is in the *proof* of the inverse map, after its integrality is proved; the final polynomial has no division.

## 5 Centered roots and shared positivity

Two algebraic identities lower the degrees of the projections. First, put $P = k + E(c - \eta)$ and $\tau_0 = \tau + E(c - \eta)$. Then

$$\tau_0^2 - P^2 + Pk - 1 = \tau^2 + (2\tau - k)E(c - \eta) - 1. \tag{10}$$

This integral triangular change is inherited from the recent source projections. Second, put $y_0 = V + u$ in the auxiliary norm:

$$I^2V^2 - (I^2 - 1)y_0^2 - 1 = V^2 + (1 - I^2)u(2V + u) - 1. \tag{11}$$

When $V$ is itself a polynomial, this eliminates its unnecessarily repeated leading square from the product with $I^2$.

**Lemma 5.1** (A parity-forced half-centered root)**.** *For integers* $P, k$*, integer roots of* (3) *are in bijection with integer roots of*

$$t^2 + 2t(2P - k) + k^2 - 4 = 0. \tag{12}$$

*The maps are* $t = 2\tau_0 - 2P + k$ *and* $\tau_0 = (t + 2P - k)/2$.

*Proof.* Substitution gives four times the old residual. Conversely the new residual differs from $(t - k)^2$ by $4tP - 4$. Hence a zero has $(t - k)^2 \equiv 0 \pmod 4$ and $t - k$ even. Its inverse is integral and restores the old norm. No parity promise is an added constraint. □

After $\eta = c - kY$ has been substituted, the usual centered first row has degree 24. Formula (12) has degree 18. This is useful in the intermediate twenty-variable, degree-44 unpaired form, although the shared guard dominates the degree of the eventual seventeen-variable form.

**Lemma 5.2** (Shared strict positivity; package 68)**.** *For integers $x, y$, put*

$$\mathcal{G}(x, y, s) = xy - (s^2 - x - y + 1)^2.$$

*Then $x, y > 0$ if and only if $\mathcal{G}(x, y, s) > 0$ for some integer $s$.*

*Proof.* A positive value implies $xy > 0$. If $x = -a, y = -b$ with $a, b \geq 1$, the subtracted square is at least $(a + b + 1)^2 > ab$, a contradiction. Conversely set $N = x + y - 1 \geq 1$. Then $xy \geq N$. Choose the closest integer square $s^2$ to $N$; writing $m = \lfloor \sqrt{N} \rfloor$ shows that either $s = m$ or $m + 1$ gives $|s^2 - N| \leq m$, with square error strictly below $N$ unless $N$ is square, when the error is zero. Thus $(s^2 - N)^2 < N \leq xy$. □

The two equations $G = \text{Pos}$ and $\eta\zeta = \text{Pos}$, with six private coordinates, may therefore be replaced by

$$G\eta\zeta - (s^2 - G - \eta\zeta + 1)^2 = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1), \tag{13}$$

with four private coordinates. This step is an existence equivalence, not an identity between the old and new sums of squares. Its proof and credit are retained explicitly [4].

# 6 A complete polynomial in eighteen integers

## 6.1 A transport coordinate which can be eliminated

Replace the independent coordinate $\alpha$ by $C = q - F - \alpha$. This is an integral triangular bijection, with inverse $\alpha = q - F - C$. The transport equation is now monic in $F$:

$$F = (K_0 + W)C - z(q - 1). \tag{14}$$

Substitute this expression everywhere and delete the transport row and the independent coordinate $F$. The packing index becomes

$$R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J. \tag{15}$$

These are polynomial maps in both directions on the corresponding zero sets. No divisibility, sign, rank or consistency assumption is needed for this projection. Its degree cost depends on which arithmetic coordinates have already been eliminated: here $R$ has degree five and $G$ degree four, while the leading forms of $k$ and the shared guard remain unchanged.

## 6.2 The polynomial

This section specifies the principal compact form without hidden quantifiers. Supply exactly the following eighteen integers, with no positivity restrictions:

$$j, C, z, f, h, r, v, w, \tau, \eta, u, \gamma, T, c, s, p_1, p_2, p_3. \tag{16}$$

Use the six fixed coefficients from Section 2. Define polynomial abbreviations in the following order:

$$\begin{gathered}
J = 3j + 1, \qquad q = \beta J + 1, \qquad W = 4(3w + 1), \\
F = (K_0 + W)C - z(q - 1), \quad X = qW, \quad Y = (2v + 1)q^3, \\
E = XY, \qquad a = E + Y, \qquad H = 4a + 3, \qquad \Delta = a^2 + H, \\
R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
k = R + 1 + hE, \qquad \zeta = k - \eta, \qquad I = (2r + 1)c^2, \\
V = fcT - f^2R - c, \qquad D = X + ac + H\gamma, \\
G = A_F F\, A_C C - (q - A_F F - A_C C)^2, \qquad Z_\eta = \eta\zeta.
\end{gathered} \tag{17}$$

The six residuals are

$$\begin{aligned} r_c &= c - kY - \eta, \\ r_0 &= \tau^2 + (2\tau - k)E(c - \eta) - 1, \\ r_m &= D^2 - \Delta c^2 - 1, \\ r_s &= I^2 - \Delta(f^2 - 1), \\ r_a &= V^2 + (1 - I^2)u(2V + u) - 1, \\ r_g &= GZ_\eta - (s^2 - G - Z_\eta + 1)^2 - 1 - p_1^2 - p_2^2 - p_3(p_3 + 1). \end{aligned} \tag{18}$$

The requested single Diophantine equation is

$$\boxed{P_{18} = r_c^2 + r_0^2 + r_m^2 + r_s^2 + r_a^2 + r_g^2 = 0.} \tag{19}$$

All exponents are fixed. There are no variable powers, rational coefficients, implicit sign conditions or additional quantified registers. The JSON witness names `J3`, `C_native`, `v_scale`, `w3`, `y_gap` and `pair_root` correspond to $j, C, v, w, u, s$ here.

**Theorem 6.1.** *The polynomial* (19) *has exact total degree 32. With the fixed coefficients above, it has an integer zero if and only if the specified ZFC calculus has a finite refutation. In particular ZFC proves that it has no integer zero if and only if* Con(ZFC)*.*

*Soundness.* A zero makes all six residuals vanish. Lemma 5.2 applied to $r_g$ gives $G > 0$ and $\eta\zeta > 0$. The determinant lemma restores the two earlier common-sign guards. Restore the polynomial registers in (17), $\alpha = q - F - C$, the old first root $\tau_0 = \tau + E(c - \eta)$, and $y_0 = V + u$. Then $r_c$ supplies the original $c = kY + \eta$, and every preliminary signed source equation is present. Its early strong-rank conclusion (7) gives $c, f \neq 0$ and $c \mid f^2 - 1$ without using either missing quotient. Lemma 4.1 now restores the integer quotients $o, j_{\mathrm{aux}}$ in (9). Equations (10)–(11) recover all original norms, and (14) supplies the original transport equation identically. Thus the guarded, residue-oriented source is fully restored. Its signed power and native calibration theorems yield a finite refutation. □

*Completeness.* From a finite refutation choose the common canonical source witnesses of Section 3, including both affine residue restrictions and $G > 0$. Compress the two source quotients by Lemma 4.1. Project all monically defined arithmetic coordinates, use (14), and choose the integral centered roots. Since $G$ and $\eta\zeta$ are positive, Lemma 5.2 and the ternary Pos representation supply $s, p_1, p_2, p_3$. Every row of (18) vanishes. All choices are finite arithmetic constructions from a hypothetical refutation; none assumes such a refutation actually exists. □

*Exact degree.* The leading forms in the input degree are

$$q_1 = 3\beta j, \quad X_2 = 36\beta j w, \quad Y_4 = 54\beta^3 j^3 v, \quad E_6 = 1944\beta^4 j^4 v w.$$

Thus $[k]_7 = hE_6$ and $[Z_\eta]_8 = \eta h E_6$. The guard has degree 16, with leading form $-\eta^2 h^2 E_6^2$. The other five residuals have degrees $11, 14, 14, 14, 14$. Consequently the degree-32 leading form of $P_{18}$ is precisely

$$\eta^4 h^4 E_6^4 = 1944^4 \beta^{16} j^{16} v^4 w^4 h^4 \eta^4,$$

whose coefficient is positive for the actual $\beta$. This is an exact-degree proof after coefficient specialization. □

### 6.3 A seventeen-variable equation of degree 88

Delete $\eta$ from (16), define $\eta = c - kY$, and delete $r_c$. Put $P = k + E(c - \eta)$ and replace $r_0$ by

$$r_0^{\mathrm{half}} = \tau^2 + 2\tau(2P - k) + k^2 - 4.$$

Keep the other four rows. The sum of squares of these five rows is an explicit equation $P_{17} = 0$ in seventeen unrestricted integers. Lemma 5.1 supplies the integral inverse root, so the proof of Theorem 6.1 applies.

The leading forms are now

$$[\eta]_{11} = -hE_6Y_4, \quad [\zeta]_{11} = hE_6Y_4, \quad [Z_\eta]_{22} = -h^2E_6^2Y_4^2.$$

The shared guard has degree 44 with leading form $-h^4E_6^4Y_4^4$. Its square has degree 88 and positive leading coefficient. All other rows have degree at most 18. Hence the claimed degree is exact. The gain in variables comes with a higher degree; neither $P_{17}$ nor $P_{18}$ dominates the other in both measures.

### 6.4 Twenty variables at degree twenty

There is a particularly useful alternative between the lower-degree forms and $P_{18}$. Add two independent integer coordinates $E$ and $R_*$ to the eighteen coordinates in (16). Let $R_{\mathrm{native}}$ denote the polynomial on the right of (15). Add the two residuals

$$r_R = R_* - R_{\mathrm{native}}, \qquad r_E = E - XY.$$

In every other abbreviation use $R = R_*$, keep $E$ independent, and set $a = E + Y$ as before. Take the sum of squares of these two rows and the six rows in (18). This defines $P_{20}$ in twenty integers. The two new rows uniquely restore $E$ and $R$, so the zero sets are polynomially isomorphic after forgetting these two coordinates. The consistency equivalence follows immediately from Theorem 6.1.

The residual degrees are now $5, 6, 6, 4, 10, 10, 10, 8$, in the order $r_R, r_E, r_c, r_0, r_m, r_s, r_a, r_g$. In particular $k$ has degree two and $V$ degree three. The degree-ten part of the auxiliary row is

$$[r_a]_{10} = -8r^2c^5ufT + 8r^2c^4uf^2R_*.$$

No other row involves both $r$ and $u$. Thus the coefficient of $r^4c^8u^2f^4R_*^2$ in $P_{20}$ is exactly 64, proving exact degree 20 independently of the large fixed coefficients. This illustrates why choosing *which* polynomial coordinate to retain matters: retaining the native index $R$ is better here than retaining the auxiliary $V$.

## 7 The rest of the integer tradeoff family

The exact packets include intermediate forms so that the reductions can be inspected locally. From the 37-variable quartic, the six monic eliminations

$$P = k + E(c - \eta), \quad N = q(L + 1), \quad X = qW, \quad C_2 = c^2, \quad F_2 = f^2, \quad g = y_0^2 - V^2$$

together with (10) give 31 variables and cubic residuals. Eliminating eleven lift coordinates in total from the 37-variable source, including $L = q^2-1$, $N = q^3$, $B_{\mathrm{aux}} = I^2$, $D$, $\Delta$, and $\zeta = 1-V+cj+hE-\eta$, gives 26 variables and quartic residuals. The defining rows vanish identically under the supplied restoration maps. Their inverse simply restores each deleted coordinate.

In that 26-variable system, set $V = fo - c$ and $y_0 = V + u$. The auxiliary row becomes (11), of degree five; every other remaining row has degree at most four. There are 25 variables. The two sign arguments have degrees at most two and three, so shared positivity has degree six and removes two private coordinates. This gives the 23-integer, degree-12 form.

For the CRT family it is advantageous to restore $\zeta = R + 1 + hE - \eta$ from the packing-index row instead of the auxiliary quotient row. Retain $V$ initially, impose $V - fcT + f^2R + c = 0$, and substitute $I = (2r + 1)c^2$. Before sharing the sign guards there are 24 variables and eleven rows, with maximum residual degree eight. Then proceed as follows.

| Unpaired source change | Variables | Residuals | SOS degree |
|---|---|---|---|
| Retain $V, E, a$ | 24 | 11 | 16 |
| Eliminate $V = fcT - f^2R - c$ | 23 | 10 | 26 |
| Eliminate $a = E + (2v + 1)q^3$ | 22 | 9 | 26 |
| Eliminate $E = XY$ | 21 | 8 | 28 |
| Eliminate $\eta = c - kY$ and half-center | 20 | 7 | 44 |

Sharing the two remaining sign guards changes these to 22/20, 21/26, 20/26, 19/32, and 18/88 in variables/degree. The 21/26 form is dominated by 20/26 and is not a primary release target. The other listed target packets are included, except that the degree-88 endpoint is emitted directly in its smaller transport-projected form.

Finally apply the transport projection (14)–(15). It sends 25/10 to 24/10, 23/12 to 22/16, 20/26 to 19/28, 19/32 to 18/32, and 18/88 to 17/88. Each deletes exactly one coordinate and one row. The 23/12 form remains a useful tradeoff because projection increases its degree. The 20/26 form is retained as a derivation step and is dominated by the rank-register construction $P_{20}$ above. The compact packets in `transport/` retain ordered polynomial abbreviations rather than expanding a large guard needlessly; these are ordinary fixed-degree polynomials, with no additional unknowns.

When $a$ is retained, the literal builder uses the lower-degree rows $E(X+1)-Xa$ and $c-\eta-k(a-E)$, together with $a - E - Y$. After $a$ is eliminated they become $E - XY$ and $c - \eta - kY$. These are exact row operations because

$$E(X + 1) - Xa = (E - XY) - X(a - E - Y),$$
$$c - \eta - k(a - E) = (c - \eta - kY) - k(a - E - Y).$$

There is no unsupported substitution of a rational quotient.

The highest-degree certificates are also transparent. Before deleting $V$, the centered auxiliary row has leading term $-4r^2c^4u^2$ among its degree-eight terms. After deleting $V$, its degree-thirteen pure-$j$ term is $648\beta^4r^2c^4uf^2j^4$. Its square has coefficient $419904\beta^8 > 0$. This persists when $a$ is eliminated, proving exact degree 26 for the twenty-variable shared-guard endpoint. For every expanded packet the independent finite audit additionally certifies a concrete nonzero specialized leading coefficient modulo 1,000,000,007. For the transport packets it certifies the leading coefficient of a specified univariate specialization, with every actual fixed coefficient reduced modulo the same prime. A second algorithm verifies this coefficient using finite differences and the proved degree upper bound. Such a nonzero modular residue is an exact nonvanishing certificate, not a randomized numerical estimate.

# 8 Rational saturation and separated transport weights

The two positive-domain companions below improve the wave-10 snapshot at eleven and twelve unknowns. They use the complete compact source-44 tuple

$$(\beta, K_0, M_C, M_F), \qquad \beta = 2^{590069} - 1,$$

with the already shifted high mask. The tuple must not be mixed with the source-34 tuple used by some integer-domain constructions. Its inherited native compiler and syntactic-refutation calibration remain dependencies. The new work is an exact existence reduction on the normalized positive Pell source, rather than a new claim about those compiler theorems.

The changes compose three mechanisms which were separate in wave 10: source 69's rational Pell saturation, source 70's three-root transport, and source 74's exact treatment of the possible common factor three in the adaptive main modulus. A fourth observation separates a polynomial which bounds signed roots from the multiplier which provides a positive inverse transport coordinate. They need not be the same polynomial.

| Equation | Positive unknowns | Exact degree | Gates | System gates |
|---|---|---|---|---|
| Source 70 | 11 | 325 | 113 | 110 |
| $P_{11}^{\mathrm{new}}$ | 11 | 269 | 111 | 108 |
| Source 69 | 12 | 82 | 102 | 96 |
| $P_{12}^{\mathrm{new}}$ | 12 | 78 | 102 | 96 |

The eleven-variable result reduces degree by 56/325, approximately 17.23%, and removes two arithmetic gates. The twelve-variable result reduces degree by 4/82, approximately 4.88%, at unchanged cost. Every live binary addition, subtraction, and multiplication counts as one gate, including multiplication by a fixed integer. Polynomial register names do not add quantified variables.

## 8.1 Rational Pell saturation

We reproduce the elementary source-69 theorem because it removes an entire divisibility restriction from the eleven-variable construction.

**Lemma 8.1** (Rational-coordinate saturation)**.** *If $Q > 1$ and $V$ are integers and $y$ is rational, then*

$$QV^2 - (Q-1)y^2 = 1 \quad \Longrightarrow \quad y \in \mathbb{Z}.$$

*Proof.* The equation excludes $V = 0$ and $y = 0$. Change signs to assume both are positive. Write $y = a/m$ in lowest terms, $m > 0$. Clearing denominators shows $m^2 \mid Q - 1$, so $(Q-1)y$ is integral. The map

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV$$

preserves the equation and makes $V'$ integral. When $V > 1$, the identities

$$y^2 - V^2 = \frac{V^2 - 1}{Q-1} > 0, \qquad (QV)^2 - ((Q-1)y)^2 = QV^2 + Q - 1 > 0$$

give $-V < V' < V$. The norm excludes $V' = 0$. Moreover, $Q \equiv 1 \pmod{m}$ implies that the numerator of $y'$ over denominator $m$ is congruent to $a$ modulo $m$. Its reduced denominator is still exactly $m$. Replace the new coordinates by their absolute values and repeat. The positive integral first coordinate strictly decreases until it is one, where the equation forces $y^2 = 1$ and hence $m = 1$. Since the reduced denominator was preserved, the initial $y$ was integral. □

This is an unbounded descent argument. It requires $Q$ and $V$ integral; it cannot be replaced by a claim that every rational Pell coordinate is integral. For example, $2 \cdot 1^2 - 4 \cdot (1/2)^2 = 1$ shows why the adjacent coefficients $Q$ and $Q - 1$ matter.

## 8.2 The weighted source interface and its adaptive modulus

The inherited normalized source supplies the following registers:

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, & E &= XY, \\ k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\ H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, & \ell &= EkY, \\ R &= ((q-1)(q-F) + \alpha)(q^2-1) + (M_C + qM_F)J. \end{aligned}$$

Here the supplied witnesses are strictly positive. The two scale divisibilities and two deleted quotient conditions are restored by carriers. Given any positive polynomial value $h$, define

$$\begin{aligned} W &= 1 + (q-1)h, & M &= q^3(q-1), \\ B_t &= (qK_0 + X)C - qF, \\ Z &= M(\ell + k + r) + (q-1)Y - q^3 W B_t, \\ A_0 &= Z^2 - M^2\{\ell(\ell+k) + 1\}, \end{aligned} \tag{20}$$

$$\begin{aligned} S &= qY, & U &= qS = q^2 Y, & v &= k - R - 1, \\ D_* &= U(X + ac) + H(U\gamma + SX - v), \\ A_1 &= D_*^2 - U^2(1 + \Delta c^2). \end{aligned} \tag{21}$$

The other source equations are the normalized strong and auxiliary norms

$$\begin{aligned} f^2 - \Delta i^2 c^4 &= 1, \\ QV^2 - (Q-1)y^2 &= 1, \qquad Q = \Delta(f^2-1), \quad V = f^2(cT - R) - c. \end{aligned} \tag{22}$$

The signed auxiliary coordinate is harmless; after recovery its positive gap is $|y| - V > 0$. The strong root is replaced by its absolute value when needed.

For clarity, the exact index-recovery argument using the smaller modulus $U = q^2 Y$ is recalled here. From $A_0 = 0$, the rational root $Z/M$ has an integer square and is integral. Reduction modulo $q^3$ and $q - 1$ gives

$$q^3 \mid Y, \quad q - 1 \mid B_t, \quad \tau = \ell + k + r + s - Wt, \quad s = Y/q^3 > 0, \quad t = B_t/(q-1).$$

Since $|\tau| < \ell + k$, we have $Wt > r + s > 0$. Thus $B_t > 0$ and $C > 0$. The authentic mask bounds now give $q \geq 16$ and $3q < R < q^4 - 2$. The main numerator is positive, since $D_* > Uac - Hk > k(UaY - H) > 0$. Rational-square integrality applied to $A_1 = 0$ gives a positive integral main Pell root.

Write $A = a + 2$ and $P_0 = 2XY^2 + 1$. The inherited Pell classification gives indices $n_0, p$ with

$$|\tau| = \chi_{P_0}(n_0), \quad k = 2\psi_{P_0}(n_0), \qquad D_*/U = \chi_A(p), \quad c = \psi_A(p).$$

It gives $n_0 < p$, $n_0 \geq 2$, $c > 8XY^3 \geq 8q^9 > 2R$, and $c > 2p$. The normalized strong norm makes $f = \chi_A(m)$ with $c \mid m$ and $m > 2p$. For example, strong divisibility first gives $p \mid m$; writing $m = pv_0$ and reducing $\psi_{\chi_A(p)}(v_0)$ modulo $c$ shows $c \mid v_0$ from $c^2 \mid ic^2$. The auxiliary congruences

$V \equiv -c \pmod f$ and $V \equiv -R \pmod c$, together with its norm, then invoke the inherited signed auxiliary-index lemma to yield $p = R$. None of these steps assumes the deleted scale $q \mid X$.

In both new constructions $J \equiv 1 \pmod 3$; since $\beta \equiv 1 \pmod 3$, $q \equiv 2 \pmod 3$. Therefore $\gcd(q, H) = 1$ and

$$g := \gcd(H, U) = \gcd(3, Y) \in \{1, 3\}.$$

Crucially, $q^3 \mid Y$ and $\gcd(q, g) = 1$ imply $q^3 \mid Y/g$. Hence

$$\mu = \frac{qY}{g} \geq q^4 > R.$$

The congruence $U \mid H(SX - v)$ gives $\mu \mid v$. The first recurrence gives $k \equiv 2n_0 \pmod{4XY^2}$, and $\mu \mid 4XY^2$. Thus $\mu \mid 2n_0 - R - 1$. The strict index bounds force $2n_0 = R + 1$. Consequently $v = k - 2n_0$ is a positive multiple of $4XY^2$, and so of $U$. Only now does the main congruence give $q \mid HX$, and then $q \mid X$. The recovered source coordinates are

$$w = X/q > 0, \quad h_{\text{index}} = v/(XY) > 0, \quad \gamma_{\text{old}} = \gamma + w - v/U > 0.$$

The last inequality follows from $H\gamma_{\text{old}} = D_*/U - ac - X > c - X > 0$. The first carrier restores the positive transport quotient. This is the same normalized compiler source as in the wave-10 packages.

Conversely its canonical witnesses have $q = B^N$, $B = \beta + 1$, with odd $N > 1$, $R \equiv 3 \pmod 4$, $X = 2^R$, $q^3 \mid Y$, and

$$t_0 = B_t/(q - 1) = q(z_{\text{source}} - 1) \geq q, \quad s = Y/q^3 < k, \quad \tau > \ell.$$

For $h \geq k$ the choice

$$r = Wt_0 + \tau - \ell - k - s > qW - 2k > 0$$

restores the first carrier. The main coordinate $\gamma = \gamma_{\text{old}} - w + v/U$ is positive, because $v/U \geq 4XY/q^2 > w$. These are the precise completeness facts used below.

### 8.3 The eleven-variable polynomial

Supply exactly

$$j, F, \alpha, i, T, X, Y, n, \eta, \zeta, \gamma \in \mathbb{Z}_{>0}, \qquad J = 3j + 1.$$

Use the common registers above and put

$$\begin{aligned}
&I = ic^2, && D_f = 1 + \Delta I^2, && Q = \Delta(D_f - 1),\\
&\delta = Q - 1, && V = D_f(cT - R) - c, && D_y^{\text{num}} = QV^2 - 1,\\
&\widehat{R} = q^3(q + F + \alpha), && \overline{V} = D_f(cT + \widehat{R}), && L = 3\overline{V},\\
&h = D_f cT, && W = 1 + (q - 1)h, &&\\
&\overline{Z} = M(\ell + k + n + L) + (q - 1)Y - q^3 W B_t. &&&&
\end{aligned} \tag{23}$$

Use $A_1$ from (21) and define the octic norm schedule

$$\begin{aligned}
&\mathcal{A} = M^2\{\ell(\ell + k) + 1\}, && \mathcal{B} = M^2 D_f, && \mathcal{C} = M^2 D_y^{\text{num}},\\
&T_n = \delta^2 \mathcal{A}\mathcal{B}, && R_n = \delta\mathcal{C}, &&\\
&J_n = \delta(\overline{Z}^2 - \mathcal{A} - \mathcal{B}) - \mathcal{C}, &&&&\\
&N_* = J_n^2 - 4\{T_n + R_n(\mathcal{A} + \mathcal{B})\}, &&&&\\
&N_8 = N_*^2 - 64 T_n R_n \overline{Z}^2. &&&&
\end{aligned} \tag{24}$$

The requested ordinary polynomial is

$$\boxed{P_{11}^{\mathrm{new}} = N_8 + A_1 + N_8 A_1.} \tag{25}$$

There are no divisions in this definition. Its literal evaluator is `p11_positive_degree269.json`.

**Theorem 8.2.** *The polynomial (25) has a strictly positive zero exactly when the inherited normalized compiler source has a positive solution.*

*Proof.* The equation is $(1 + N_8)(1 + A_1) = 1$. As $N_8$ is a square modulo four, $N_8 \neq -2$. The two integer factors therefore must both be one. Thus the single equation is equivalent to $N_8 = A_1 = 0$. No parity restriction on $A_1$ is needed for this two-factor step.

For the root extraction write $D_\tau = \ell(\ell + k) + 1$ and $D_y = D_y^{\mathrm{num}}/\delta$ in the proof only. The standard eight-conjugate identity gives

$$N_8 = \delta^4 \prod_{\epsilon_1,\epsilon_2,\epsilon_3=\pm 1} \left(\overline{Z} - M\epsilon_1\sqrt{D_\tau} - M\epsilon_2\sqrt{D_f} - M\epsilon_3\sqrt{D_y}\right).$$

Its radicands are positive before any equation is imposed. Indeed $1 < D_\tau < D_f < Q$; also, writing $b = cT - R \in \mathbb{Z}$, either $b = 0$ and $V = -c$, or $|V| \geq D_f - c > 1$. Therefore

$$D_y = V^2 + \frac{V^2 - 1}{Q - 1} > V^2 > 1.$$

It cannot equal $D_\tau$ or $D_f$: equality to an integer $D$ with $1 < D < Q$ would give $Q(D - V^2) = D - 1$, an impossibility. Thus the three radicands are distinct.

Their signed sum cannot be zero. With positive roots $t, f, y$, one has $t < (a+1)c$ and $f > (a+1)c^2$. If $b = 0$, then $y < c + 1$ and $f > t + y$. If $b \neq 0$, then $y > |V| \geq f^2 - c > 2f > f + t$. Consequently a zero of the norm supplies signed roots with nonzero rational sum

$$\overline{Z}/M = \tau_* + f_* + y_*.$$

For completeness, a nonzero rational sum of three signed roots of distinct positive rational numbers forces all three roots rational. To see this, let $d_i = x_i^2$ and $t = x_1 + x_2 + x_3 \neq 0$. Then

$$u = \frac{t^2 - d_1 - d_2 - d_3}{2}, \qquad v = \frac{u^2 - d_1d_2 - d_1d_3 - d_2d_3}{2t}$$

are rational and equal respectively to the pairwise sum and product of the roots. The identities

$$d_1t + v = x_1(x_1 + x_2)(x_1 + x_3), \qquad x_1 = \frac{d_1(d_1 + u)}{d_1t + v}$$

recover $x_1$ rationally. The denominator is nonzero by distinctness of the positive $d_i$, and the cyclic formulas recover the other roots.

Since $D_\tau, D_f$ are integers, $\tau_*, f_*$ are integers. The third root satisfies $QV^2 - (Q-1)y_*^2 = 1$, with integral $Q > 1$ and $V$. Lemma 8.1 makes $y_*$ integral as well. This replaces source 70's coprimality argument and permits $I = ic^2$ throughout.

The authentic mask inequalities $0 < M_C < \beta$, $0 < M_F < 2\beta$ imply $|R| < \widehat{R}$. Thus $|V| < \overline{V}$, $f < \overline{V}$, and $y < 2|V| < 2\overline{V}$, so $L > f + y$. Set

$$r = n + L - f_* - y_* > 0, \quad f_{\mathrm{old}} = |f_*|, \quad y_{\mathrm{old}} = |y_*|, \quad u = y_{\mathrm{old}} - V > 0.$$

The first carrier becomes $Z = \overline{Z} - M(f_* + y_*) = M\tau_*$. The strong and auxiliary source norms hold by construction. All the weighted source witnesses are now positive, and $h = D_f cT > 0$. The preceding adaptive argument restores the normalized source.

Conversely choose canonical normalized source witnesses and its positive auxiliary reconstruction, with $f^2 = D_f$, $y = V + u > 0$. The latter can be constructed from the odd index $R \equiv 3 \pmod 4$ using $y = \psi_{\Delta I}(R)$ and $V = \chi_{\Delta I}(R)/(\Delta I)$. The squared-modulus congruences make $T = (V + c + Rf^2)/(cf^2)$ a positive integer. No extra factor $q(q-1)$ in $I$ is required. Here $F + \alpha < q$ and $c > 8q^9$, so

$$\widehat{R} < 2q^4 < c \le cT, \qquad L < 6D_f cT = 6h.$$

Also $h > k$. For the canonical first carrier,

$$r > qW - 2k > q + \{q(q-1) - 2\}h > 6h > L.$$

Thus $n = r - L + f + y > 0$, and substitution gives $\overline{Z} = M(\tau + f + y)$, so $N_8 = 0$. The adaptive main coordinate is positive by the earlier completeness argument. Finally the canonical repunit has $J \equiv 1 \pmod 3$ and $J > 1$, so $j = (J-1)/3 > 0$. □

### 8.4 The twelve-variable polynomial

Supply exactly

$$j, F, \alpha, t, T, X, Y, r, \eta, \zeta, n, \gamma \in \mathbb{Z}_{>0}, \qquad J = 6j + 1.$$

Use (20) with $h = k$ and (21). Put

$$\begin{aligned}
&f = t + 1, && \rho = f^2 - 1, && d = \Delta c^4,\\
&Q = \Delta\rho, && V = f^2(cT - R) - c, &&\\
&a_* = (V+n)^2 - Q\{(V+n)^2 - V^2\} - 1, &&&&\\
&b_* = 2(1-Q)(V+n), && A_* = da_* + (1-Q)\rho, &&\\
&N = A_*^2 - db_*^2\rho. &&&&
\end{aligned} \tag{26}$$

Define

$$\boxed{P_{12}^{\mathrm{new}} = (1 + A_0)(1 + A_1)(1 + N) - 1.} \tag{27}$$

This is precisely source 69's saturated merger with the smaller adaptive main modulus. The literal evaluator is `p12_positive_degree78.json`.

**Theorem 8.3.** *The polynomial (27) has a strictly positive zero exactly when the inherited normalized compiler source has a positive solution.*

*Proof.* Here $q$ is even, since $\beta$ and $J$ are odd. Therefore $A_0$ and $A_1$ are squares modulo four; so is $N$, because $b_*$ is even. The product equals one exactly when all three residuals vanish.

The first and main carriers recover $C > 0$ and $c > 2R$ before the strong norm is used. Since $f \ge 2$ and $T \ge 1$, $V = f^2(cT - R) - c > c > 1$. Thus $b_* \ne 0$ at a candidate zero. Define rational candidates $i = A_*/(db_*)$ and $u = n - i$. The norm gives $di^2 = \rho$. If

$$H_0(v) = (1-Q)v^2 + 2(1-Q)Vv + V^2 - 1,$$

then direct expansion gives $H_0(n - i) = 0$. Hence $y = V + u$ satisfies $QV^2 - (Q-1)y^2 = 1$. Lemma 8.1 gives $y, u, i \in \mathbb{Z}$. Replace $i$ by $|i| > 0$ and the gap by $|y| - V > 0$; the positivity of the

latter follows from $y^2 - V^2 = (V^2 - 1)/(Q - 1) > 0$. These are the normalized strong and auxiliary source witnesses, and the adaptive carrier restores the remaining source quotients.

Conversely, canonical normalized witnesses supply positive $f, i, u$; take $t = f - 1 > 0$ and $n = i + u > 0$. The identity

$$N|_{\rho=di^2} = d^2 H_0(n - i)H_0(n + i)$$

gives $N = 0$. The canonical first and adaptive main carriers vanish by the preceding reconstruction. The repunit has $J \equiv 1 \pmod 6$ and $J > 1$, so $j = (J - 1)/6 > 0$. This proves completeness. □

### 8.5 Exact degrees, coefficient certificates, and audit scope

Write $q_0 = 3\beta j$ for the eleven-variable polynomial and $q_0 = 6\beta j$ for the twelve-variable one. In either case put

$$k_0 = \eta + \zeta, \quad R_4 = q_0^3(q_0 - F), \quad B_{t,2} = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F.$$

The cancellation in the main Pell norm gives

$$[A_1]_{13} = 8q_0^2 k_0 X^2 Y^4 (q_0^2 Y\gamma + R_4).$$

For eleven variables, the exact degrees of the principal registers are

$$\deg I = 5, \quad \deg D_f = 14, \quad \deg Q = 18, \quad \deg h = 17, \quad \deg L = 18, \quad \deg \overline{Z} = 23.$$

If $D_{f,14} = i^2 k_0^4 X^2 Y^6$ and $Q_{18} = i^2 k_0^4 X^4 Y^8$, then

$$[\overline{Z}]_{23} = -q_0^4 D_{f,14} k_0 Y T B_{t,2}, \qquad [N_8]_{256} = Q_{18}^4 [\overline{Z}]_{23}^8.$$

Multiplication by the displayed nonzero degree-13 main leader proves the upper bound 269 is attained. More specifically, the coefficient of

$$j^{36} i^{24} \eta^{57} X^{42} Y^{93} T^8 \alpha^8 \gamma$$

is exactly $8(3\beta)^{36} > 0$. The value is independent of $K_0, M_C, M_F$, so it remains nonzero after their actual fixed specialization.

For twelve variables the residual degrees are $15, 13, 50$. Their leading forms are the main leader above together with

$$[A_0]_{15} = -2q_0^8 k_0^2 X Y^2 B_{t,2}, \qquad [N]_{50} = 4k_0^8 X^8 Y^{16} t^8 R_4^2 n^2.$$

The coefficient of $j^{18} F^2 \eta^{11} X^{12} Y^{23} t^8 n^2 \gamma\alpha$ in $P_{12}^{\mathrm{new}}$ is $64(6\beta)^{18} > 0$. Thus its degree is exactly 78.

The included standard-library audits independently expand the literal polynomial formulas, prove the universal norm and descent identities, check all live inputs and gates, and compute these exact leading forms. The twelve-variable audit checks 13,128 terms in the first carrier and 4,417 in the main carrier; it handles the merger as an exact 32-term polynomial cut. These finite computations do not decide consistency or replace the unbounded proofs. Composing the two existence equivalences with the fixed source calibration, and negating, yields the corresponding ZFC-provable no-positive-zero equivalences with Con(ZFC).

# 9 A majorant for the complete Pell radical

A separate construction improves the lowest inspected degree with nine strictly positive unknowns. Its degree is

$$\boxed{1\,158\,228},$$

compared with package 71's 1 379 730, a reduction of 16.0540% [7]. Its delivered evaluator uses 726 binary operations, compared with 478; this is a degree–size tradeoff. All four coefficient payloads are the coherent original31 tuple. The new polynomial does not add a witness or change the native compiler.

## 9.1 The complete radical and the denominator that can be cleared

For $A \geq 3$, $C, i \geq 1$, and $t \geq 3$, put

$$\begin{aligned} D &= (A^2-1)C^2+1, & e &= i(4A-5)C,\\ F &= (D-1)e^2D^2+1, & L &= CD+1-A,\\ G &= A+FL, & H &= C+Ft,\\ I &= (G^2-1)H^2+1. \end{aligned}$$

All these registers are positive, and $G > 1$. Regard $A$ as a formal indeterminate over $\mathbb{Q}(C, i, t)$. The radical $\sqrt{DFI}$ has a unique formal Laurent expansion with leading term $2^{10}C^{24}i^5tA^{23}$. Its polynomial part provides the construction

$$T_0 = 2^{16}C^7[A^{\geq 0}]\sqrt{DFI}. \tag{28}$$

The expression $T_0$ is an integer polynomial with 654 monomials. Its complete table is the `majorant` field of `nine_full_radical/laurent_candidate.json`; the coordinate order is $(A, C, i, t)$. This table is an ordinary division-free definition. Formula (28) explains how it was discovered.

The exact polynomial

$$\Delta_0 = T_0^2 - 2^{32}C^{14}DFI \tag{29}$$

has $A$-degree 22. Together with the leading term of $T_0$, this also proves the formal-polynomial-part assertion: dividing the difference of the squares by their formal sum leaves only negative powers of $A$. The factor $C^7$ clears variable denominators without raising the $A$-degree.

**Complete positivity certificate.** Under

$$A = x+3, \qquad C = y+1, \qquad i = z+1, \qquad t = u+3,$$

$T_0$ has exactly 9 216 nonzero coefficients and $\Delta_0$ exactly 32 637. Every one is a positive integer, including the constant coefficients. The archive contains both complete expansions and an independent verifier that reconstructs the radical and checks every coefficient. Hence, throughout the stated real domain,

$$T_0 > 2^{16}C^7\sqrt{DFI} > 0. \tag{30}$$

This is a finite algebraic proof. No tests on a bounded selection of integer inputs replace the coefficient identities.

The improvement comes from approximating the whole radical. The preceding seventh-order construction approximated $\sqrt{DF}\,G$ and multiplied the error by the high-degree factor $H$. Taking the polynomial part of the complete radical prevents that multiplication of the approximation error.

### 9.2 The final polynomial and its correctness

Retain the source and Pell registers of package 71, with positive unknowns

$$b, g, h, s, w, C, i, j, n.$$

The source has convolution height 1873 and maximum coordinate weight 116. Its full literal circuit, all kernels, and the fixed coefficient generator and compressed integer payloads are included in the companion directory. In particular,

$$\begin{gathered}
U = N^2 w, \quad Y = N^2 s, \quad M = RY, \quad P = 2M^2 U, \\
K = R + 1 + h(P-1), \quad A = M(U+1), \quad B_1 = 2R+1, \quad W_1 = bw, \\
J = 4A - 5, \quad Q = 3W_1 C - 2(W_1^2 - 1), \quad t = B_1 + (j-1)C, \\
a_0 = (P^2-1)K^2 + 1, \quad V_0 = PK, \\
m = K^2(4\,\mathrm{cap} - g) - 16\,\mathrm{cap}(C - KY)^2, \\
\Omega = Jm, \qquad E_0 = J(n + W_1^2) + Q.
\end{gathered}$$

Define

$$\begin{gathered}
S = 2^{18} C^7, \qquad T = 4T_0(A, C, i, t), \qquad a_1 = S^2 DFI, \\
\Theta = E_0 - \Omega(V_0 + T), \\
\boxed{P_9 = (\Theta^2 - \Omega^2(a_0 + a_1))^2 - 4\Omega^4 a_0 a_1.}
\end{gathered}$$

All displayed registers are abbreviations, not additional unknowns. The authoritative evaluator is `nine_full_radical/laurent9_positive.json`.

The unchanged source estimates give $R \geq 1$, $A \geq 3$, and hence $t \geq 3$, unconditionally on positive inputs. Thus $T > \sqrt{a_1}$ before any source zero has been decoded. Also $V_0^2 - a_0 = K^2 - 1 > 0$. If $m = 0$ the norm is $E_0^4 > 0$; otherwise its real factorization gives signed roots $u_0^2 = a_0$, $u_1^2 = a_1$ and

$$E_0/J = m(V_0 + T + u_0 + u_1).$$

The parenthesis is strictly positive. Therefore $m > 0$ before the square-root integrality argument. The inherited margin argument gives $C > K$, $C > B_1 + W_1$, and $DFI > a_0$. In particular $a_1 > a_0$ and the rational sum $u_0 + u_1$ is nonzero. The identity

$$2(u_0 + u_1)u_0 = (u_0 + u_1)^2 + a_0 - a_1$$

makes both roots rational and then integral. Since $(u_1/S)^2 = DFI$, the rational-root argument also makes $u_1/S$ an integer. The factor equation recovers $J \mid Q$, and the existing ratio theorem supplies the encoded positive source zero.

Conversely, choose the inherited ratio witnesses from a source zero, take the positive integral roots $r_0 = \sqrt{a_0}$ and $r_1 = \sqrt{DFI}$, and set

$$n = m(V_0 + T + r_0 + Sr_1) - W_1^2 - Q/J.$$

It is an integer. The inherited bound $W_1^2 + Q/J < 2D$, together with $T > S\sqrt{DFI} > SD > 2D$ and $m \geq 1$, shows $n > 0$. One norm factor now vanishes. Thus the positive existence statements are equivalent in both directions. Combining with the specified native compiler and refutation calibration gives the desired ZFC-provable consistency equivalence. These inherited semantic results are not being presented as newly formalized proofs.

### 9.3 Exact degree and evaluator cost

For this unchanged source let

$$r = \deg R = 7963, \quad q = \deg N = 2115, \quad a = \deg A = 16425, \quad \omega = \deg \Omega = 82360.$$

The leading $T_0$ monomial is $2^{26}A^{23}C^{31}i^5t$; the highest weighted monomial of (29) is

$$141664530202624\, A^{22}C^{42}i^{10}t^2.$$

Therefore

$$\deg T = 23a + r + 36 = 385774, \qquad \deg(T^2 - a_1) = 22a + 2r + 52 = 377328.$$

Write $\delta_0 = K^2 - 1$, $\delta_1 = T^2 - a_1$, and

$$B_* = E_0^2 - 2E_0\Omega(V_0 + T) + \Omega^2(\delta_0 + \delta_1).$$

The exact identity

$$P_9 = B_*^2 + 4\Omega^2 V_0 T B_* + 4\Omega^4(\delta_0 T^2 + a_0\delta_1)$$

has four candidate degrees

$$1084096, \qquad 1149781, \qquad 1158228, \qquad 821246.$$

The unique dominant term is $4\Omega^4\delta_0 T^2$. On the positive diagonal its leading coefficient is

$$2^{166}a_b^{578828}c_b^{276} > 0, \qquad a_b = 2^{2134939}, \quad c_b = 2^{17079501}.$$

Consequently the asserted degree is attained by the actual fixed coefficient instance.

The 726-operation evaluator uses a 110-term correction to the old majorant, in the existing registers $J, C, i, t, D$:

$$T = 64C^7(T_7 - 640i - 4096)H - \mathrm{Corr}(J, C, i, t, D).$$

The correction table and the exact identity check are included. There are 430 multiplications, 265 additions, and 31 subtractions; coefficient leaves and register reuse are free under the same convention as the comparison source.

The new degree limit is now set by the first-root defect $K^2 - 1$. A tighter simultaneous bound for both norm roots is a concrete further research question, as are cheaper evaluators of the correction and an application to the ten-integer finalizer. None of those additional improvements is claimed here.

## 10 Verification, interpretation and integration

The release is designed to make each mathematical claim inspectable. The integer builder reads the source packets as inert arithmetic expressions, replaces the two guard rows, emits the projected and CRT families, and records exact counts, liveness and degree certificates. The parser accepts only integer literals, names, additions, subtractions, products and fixed nonnegative powers. An independent checker uses a separately written sparse polynomial ring and does not import the builder. It verifies the source bridge, row operations, successive projections, parity identity, factored formulas and actual coefficient specializations. A separate transport checker proves the local inverse

identities, compares every retained row and definition, and verifies degree by finite differences independently of the builder's univariate polynomial algorithm.

The coefficients are independently regenerated from the retained finite machine/tile recipe. The positive and nine-variable directories contain their own offline verifiers and coherent fixed coefficient payloads. The accompanying receipts state the checks actually performed. All new release checks use explicit failures rather than Python assertions; an optimized-Python run is also supported.

It is neither possible nor useful to validate a consistency equation by exhaustively searching for a zero. A finite-box search would say little about (1). The checks here instead establish exact finite claims: the emitted polynomial is the proved polynomial, its coefficients are the specified numerals, its transforms are algebraically correct, and its counts and leading forms are accurate. The proofs of unbounded Pell rank, native source soundness and machine/refutation calibration remain the explicitly retained mathematical inputs.

The recommended starting point for integration is the eighteen-variable equation (19): it has only six residuals, moderate degree 32, and a direct reconstruction proof. For a lower degree at the cost of one coordinate, use the nineteen-variable degree-28 form; at the cost of two, use the twenty-variable degree-20 form. For fewer integer variables, use the seventeen-variable degree-88 form. For a quartic target, use the direct 37-variable variant. There is no mathematically justified total ordering which calls one of these "the simplest" without specifying a metric.

## 11 Alternative encodings and further research

The current snapshot contains a direct syntactic proof-checker route and an executable direct quartic construction. Its native-independent internal ZFC derivation and useful measured size remain unfinished in the reviewed sources [9]. We did not treat an abstract appeal to DPRM, an unexpanded compiler call, or a finite local transition relation as a smaller completed consistency equation.

A promising separate direction replaces the many axiom-schema branches of a direct ZF/ZFC checker by a fixed finite first-order axiomatization of a suitable class theory. A conservative equiconsistency theorem, exact proof calculus, and its internal arithmetic encoding would still be required. Counting fewer local constructors alone does not measure the global polynomial: syntax validation and unbounded proof aggregation can dominate. The accompanying alternative-route note makes these remaining requirements explicit.

The following questions arise directly from the proved reductions.

1. **Formalize the quotient lemma at its actual dependency point.** Prove the preliminary $p \mid m$ interface, then $f^2 \equiv 1 \pmod{c^2}$, before importing any exact auxiliary index. This is a smaller and cleaner formal target than immediately formalizing the whole polynomial.

2. **Search for other inverses already supplied by a norm.** The CRT reduction uses $f$ as its own inverse modulo $c$. Other norm equations may supply polynomial inverses for scale or transport congruences and eliminate further quotient coordinates.

3. **Jointly optimize projection and degree.** Eliminating a monic coordinate can substantially increase the degree of a shared positivity guard. An optimizer should keep separate scores for arithmetic rows and sign rows, with verified inverse maps.

4. **Reduce the seventeen-variable guard degree.** Its degree 88 comes entirely from applying shared positivity after eliminating $\eta$. A different representation of that sign condition could improve the last tradeoff without changing any machine encoding.

5. **Lower fixed coefficient height.** The determinant proof uses special source-34 leading anchors. A genuine proof for the whole smaller compact tuple could shrink the fixed numerals; substituting unrelated ports is not justified.

6. **Improve full-radical evaluation.** The new nine-variable degree gain costs 248 additional gates relative to source 71. More factorization or a shorter certified majorant would be needed for simultaneous improvement in both measures.

7. **Complete a distinct calibration.** A direct proof-checker or finite-theory route should deliver its literal coefficients and the internal consistency biconditional, together with actual variable and operation counts. It would then be meaningfully comparable with the completed native-source equations.

The present contribution gives substantial new low-degree tradeoffs and an independently reusable quotient-elimination argument. It does not prove an absolute minimum, remove the inherited native calibration, or establish external novelty beyond the inspected repository and cited mathematics.

## References

[1] V. Reshetnikov, *ProveIt: Con(ZFC) in Diophantine form*, snapshot `cf4a5218a5a4e4358d3f5314151b6b05d4aa384e`, repository collection.

[2] ProveIt, *PROPOSAL_REVIEW_WAVE10.md*, same snapshot; retained comparison and proof-status boundary.

[3] ProveIt package 64, *Structural reductions*, `integer41/PROOF41.md`, `PROOF44.md`, and literal quartic.

[4] ProveIt package 68, *Joint positivity*, `integer_system/PAIR_GUARD_PROOF.md`.

[5] ProveIt package 69, *Cones and Pell saturation*, `positive/PROOF.md` and `integer/PROOF.md`.

[6] ProveIt package 70, *Residue orientation and root transfer*, `integer/PROOF.md`, `eleven/PROOF.md`, and literal source packets.

[7] ProveIt package 71, *Calibrated determinant guards*, `proofs/DETERMINANT_GUARD.md` and `nine_positive/PROOF.md`.

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[9] ProveIt, *ALTERNATIVE_ENCODINGS.md*, same snapshot.

[10] A. Danilkin and L. Chevalier, *Three Squares Theorem*, Archive of Formal Proofs, 2023, https://isa-afp.org/entries/Three_Squares.html.

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